SYSTEM ASSUMPTIONS

IoT Hydroponic System

1. INDOOR USE ASSUMPTIONS

1.1 Environmental Conditions

A. Temperature Range:

- Minimum ambient: 15°C (winter, unheated spaces)
- Maximum ambient: 35°C (summer, non-air-conditioned)
- Operational range: System must maintain 18-26°C internally
- Allowable fluctuation: ±3°C during control cycles

B. Humidity Considerations:

- Indoor RH range: 30-80% (coastal vs inland South Africa)
- System must compensate for external humidity variations
- No condensation on electrical components allowed

C. Air Quality:

- Dust levels: Medium (regular cleaning required)
- VOC exposure: Minimal (residential environment)
- CO2 levels: 400-1000 ppm (typical indoor range)

1.2 Space Constraints

A. Physical Dimensions:

- Maximum footprint: 0.6m × 0.4m (balcony/kitchen space)
- Height restriction: 1.2m (under counter/table compatibility)
- Weight when operational: <25kg (portability requirement)
- Clearance requirements: 10cm ventilation space on all sides

B. Access Requirements:

- Front access only for maintenance
- Top access for plant harvesting
- Side access for reservoir filling
- No permanent installation (rental-friendly)

C. Aesthetic Considerations:

- Domestic-appliance appearance
- Noise level <40dB at 1m distance
- No water spills/drips during normal operation
- LED lighting contained within unit

2. POWER AVAILABILITY ASSUMPTIONS

2.1 Grid Power Characteristics

A. Voltage Specifications:

- Primary: 220-240V AC, 50Hz
- Allowable fluctuation: 210-250V (Eskom typical)
- Surge protection required for ±500V spikes
- Brownout operation down to 180V

B. Load Shedding Projections (2027):

- Stage 1-2: 2-4 hours/day, predictable schedules
- Stage 3-4: 4-8 hours/day, possible during growing season
- Stage 5-6: Emergency planning required
- Scheduled vs unscheduled outages: 70%/30% ratio

C. Power Quality Issues:

- Frequency variation: 49.8-50.2 Hz
- Harmonic distortion: <8% THD assumed
- Voltage sags: 10-30% drops, 2-10 second duration
- Momentary interruptions: 3-5 events/month

2.2 Backup Power Strategy

A. Minimum Operational Requirements:

- Critical systems: Water circulation (30W), Control (10W)
- Non-critical: LED lighting, Active cooling
- Survival mode: Circulation 5min/hour, sensors active

B. Power Budget:

- Normal operation: 120-180W (peak)
- Emergency operation: 40-60W
- 24-hour consumption: 1.8-2.5 kWh/day
- Monthly cost: R250-R350 at R1.80/kWh (2027 estimate)

C. Backup Solutions:

- Integrated battery: 2–4-hour runtime at 60W
- UPS compatibility: Standard 600VA UPS support
- Solar option: 100W panel with MPPT (optional)
- Manual generator interface (12V input)

2.3 Power Management System

A. Tiered Power Priority:

1. Microcontroller & sensors (always powered)
2. Water circulation pump (priority during outages)
3. Environmental controls (heat/cool)
4. LED lighting (can be suspended)

B. Outage Response Protocol:

- Pre-outage: Full reservoir, reduce LED intensity

- During outage: Enter power-saving mode

- Post-outage: Gradual restart, sensor validation

- Extended outage: Alert user, recommend manual intervention

C. 2027 Infrastructure Improvements:

- Widespread prepaid electricity systems

- Increased residential solar adoption (~15% penetration)

- Smart grid integration potential (demand response)

- Improved outage notification APIs

3. WI-FI CONNECTIVITY ASSUMPTIONS

3.1 Network Infrastructure

A. Residential Connectivity (2027 Projection):

- Urban areas: 95% Wi-Fi coverage, 50+ Mbps average

- Peri-urban: 85% coverage, 20-30 Mbps average

- Rural: 70% coverage, 10-20 Mbps average (satellite growing)

- Mobile data: 5G coverage 80% urban, 4G 95% national

B. Network Reliability:

- Uptime: 99% urban, 95% rural (excluding load shedding)

- Latency: <50ms urban, <100ms rural

- Packet loss: <1% urban, <3% rural

- Monthly data: 50-100GB typical household

C. Cost Structure:

- Fixed-line: R500-R800/month uncapped
- Mobile data: R150-R300/50GB
- IoT-specific plans: R50-R100/month (emerging market)

3.2 IoT Communication Requirements

A. Data Volume Estimates:

- Sensor data: 2-5KB/hour (compressed)
- Camera images: 50-200KB each (optional)
- Firmware updates: 1-5MB (monthly)
- Total monthly: 10-50MB without camera, 500MB-2GB with

B. Communication Protocol Stack:

- Primary: Wi-Fi 802.11n/ac (2.4GHz compatible)
- Fallback: Bluetooth LE for local configuration
- Emergency: GSM/LTE module (optional add-on)
- Local storage: 7-30 days onboard (SD card)

C. Cloud Service Assumptions:

- AWS/Azure/GCP available in South Africa (Johannesburg region)
- Local IoT platforms: Vodacom IoT, MTN IoT, Flickswitch
- Data sovereignty: GDPR-like regulations in effect
- API availability: 99.5% uptime SLA expected

3.3 Connectivity Resilience

A. Network Failure Handling:

- Local operation: 30-day autonomy with basic functions
- Store-and-forward: Data caching during outages
- Multi-AP support: Automatic failover to backup SSID
- Mobile hotspot: Automatic detection and connection

B. Security Requirements:

- WPA3 support mandatory
- TLS 1.3 for all cloud communications
- Certificate-based authentication
- Regular security updates (automatic)

C. 2027 Technology Trends:

- Wi-Fi 6E common in urban homes
- Mesh networks in 40% of households
- IoT-specific bandwidth allocation (QoS)
- Standardized smart home protocols

4. SOUTH AFRICA-SPECIFIC ASSUMPTIONS

4.1 Economic Factors

A. Target Market Affordability:

- System cost target: R5,000-R8,000 retail
- Monthly operating cost: <R500 (including electricity)
- Payback period: 18-24 months (vs. store-bought greens)
- Target income bracket: R15,000-R40,000/month households

B. Supply Chain Considerations:

- Local component availability: 60% (by 2027)
- Import duties: 15-25% on electronic components
- Local manufacturing: Assembly possible, full production limited
- Spare parts: 3-5 day delivery in major metros

C. Market Readiness (2027):

- Hydroponic awareness: 30-40% of urban population
- Health consciousness: 50%+ prioritize fresh, local produce
- Tech adoption: 70% smartphone penetration, 60% smart home interest
- Sustainability focus: Carbon tax driving local food production

4.2 Regulatory Environment

A. Electrical Compliance:

- SANS/IEC standards mandatory
- NRCS certification required for sale
- Energy efficiency labeling (planned for 2026)
- E-waste regulations (extended producer responsibility)

B. Agricultural Regulations:

- No GMO restrictions for personal use
- Water use registration (if > specific volume)
- Nutrient solution disposal guidelines
- Biosecurity: Restricted plant species

C. Data Privacy:

- POPIA fully enforced (Protection of Personal Information Act)
- Cross-border data transfer restrictions
- Consumer data rights well-established
- Mandatory breach notification

4.3 Climate Adaptation Requirements

A. Regional Variations:

- Coastal (Cape Town/Durban): High humidity, moderate temps
- Inland (Gauteng): Low humidity, larger temp swings
- Arid (Karoo): Very low humidity, high diurnal variation
- System must auto-adapt to local conditions

B. Water Considerations:

- Municipal water: pH 7.0-8.5, hardness varies

- Load shedding impacts water pressure/pumping

- Drought restrictions possible (Level 1-3)

- Rainwater harvesting compatibility (optional)

C. Growing Season Extension:

- Year-round operation required

- Supplemental heating November-April in some regions

- Supplemental cooling September-March in others

- Seasonal adjustment algorithms needed

5. TECHNICAL INFRASTRUCTURE ASSUMPTIONS

5.1 User Technical Capability

A. Technical Literacy (2027 Urban South Africa):

- Smartphone proficiency: 85% of target market

- App installation/configuration: 70%

- Basic troubleshooting: 60%

- Network configuration: 40%

B. Support Expectations:

- Local language support: English primary, Afrikaans/Zulu available

- Support channels: WhatsApp, email, video call

- Community troubleshooting: Online forums, social groups

- Professional installation: Optional service (R500-R1000)

C. Learning Curve:

- Setup time: <2 hours for average user
- Training materials: Video tutorials, augmented reality guides
- Error recovery: Guided troubleshooting in app
- Maintenance: 30 minutes/week expected

5.2 Maintenance & Service Assumptions

A. Component Lifespan:

- Pumps: 10,000 hours (2-3 years continuous)
- LEDs: 25,000 hours (5+ years)
- Sensors: 2-3 years (pH sensors 12-18 months)
- Electronics: 5+ years (with firmware updates)

B. Consumables:

- Nutrients: R100-R200/month
- pH adjustment chemicals: R50/month
- Filter replacements: R100 every 6 months
- Growing media: R50 per cycle

C. Service Network:

- Major centers: On-site service available
- Regional areas: Courier-based replacement
- DIY repairs encouraged with parts kits
- Software updates: Automatic, free for 5 years

6. RISK MITIGATION BASED ON ASSUMPTIONS

6.1 Assumption Validation Schedule

Quarterly Review (2027):

1. Power grid stability assessment

2. Internet connectivity metrics update

3. Component availability check

4. Regulatory changes monitoring

Pre-Deployment Testing:

1. Load shedding simulation: 72-hour test

2. Network failure recovery testing

3. Temperature extreme testing (5-40°C)

4. Humidity cycling test (30-90% RH)

6.2 Contingency Plans for Invalid Assumptions

A. If Power Worse Than Expected:

- Offer integrated solar package
- Implement more aggressive power saving
- Develop manual override capabilities
- Create community charging stations concept

B. If Connectivity Insufficient:

- Enhanced local storage (90 days)
- SMS-based alert system

- Peer-to-peer mesh networking option

- Manual data sync via USB

C. If Indoor Conditions Extreme:

- Additional insulation options

- External reservoir placement

- Split system design (separate control unit)

- Climate-controlled cabinet upgrade

7. VALIDATION METHODS FOR EACH ASSUMPTION

7.1 Power Assumption Validation

Methods:

1. Historical Eskom data analysis (2024-2026)

2. Municipal power quality surveys

3. Prototype testing in actual conditions

4. Customer surveys in target areas

Metrics to Monitor:

- Outage frequency and duration

- Voltage stability measurements

- Surge event frequency

- Seasonal variations

7.2 Connectivity Validation

Methods:

1. Network speed tests in target neighbourhoods

2. ISP reliability data purchase

3. Prototype field testing with various providers

4. Mobile network coverage mapping

Metrics to Monitor:

- Signal strength in typical installations

- Data cost trends

- Latency to local cloud endpoints

- Packet loss during peak hours

7.3 User Behaviour Validation

Methods:

1. Focus groups in Johannesburg, Cape Town, Durban

2. Pilot program with 50-100 households

3. A/B testing of user interfaces

4. Ethnographic studies of potential users

Metrics to Monitor:

- Setup success rate

- Weekly interaction frequency

- Error resolution time

- Feature adoption rates

ASSUMPTION TRACKING MATRIX

Assumption Category	Confidence Level	Validation Method	Last Updated	Risk if Invalid	Mitigation Strategy
Indoor Temperature Range	High (90%)	Historical weather data	2027-Q1	System overheating	Active cooling upgrade
Power Outage Duration	Medium (70%)	Eskom forecasts	2027-Q1	Plant loss	Larger battery option
Wi-Fi Availability	High (85%)	ISP coverage maps	2027-Q1	Reduced functionality	GSM fallback module
User Tech Literacy	Medium (75%)	Market surveys	2027-Q1	Support costs	Improved onboarding
Component Costs	Low (60%)	Supplier quotes	2027-Q1	Price increase	Local sourcing options
Regulatory Compliance	High (90%)	Legal review	2027-Q1	Sales delays	Pre-certification
Water Quality	Medium (70%)	Municipal reports	2027-Q1	pH instability	Filtration system

KEY TAKEAWAYS

1. **Power Resilience is Critical**: Design must handle 4–8-hour daily outages
2. **Connectivity Should Be Robust**: Include multiple fallback options
3. **Adaptability to Local Conditions**: System must self-adjust for regional variations
4. **User Experience Must be Intuitive**: Account for varying technical skills
5. **Cost Targets Must Be Maintained**: South African market is price-sensitive
6. **Local Compliance is Non-Negotiable**: Build for SANS standards from start
7. **Climate Considerations are Regional**: Different algorithms for coastal vs inland